

SOLIDWORKS API (.NET VBA C#) — Reference

(49/337): IComponent2 (part 2 of 4)

SOLIDWORKS API Documentation (converted from CHM)

IComponent2 (part 2 of 4) (Type)

SOLIDWORKS API Help

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GetModelTexture Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetModelTexture Method (IComponent2)

Gets the texture applied to this lightweight component in the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function GetModelTexture( _  
    ByVal ConfigName As System.String _  
) As Texture
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim ConfigName As System.String  
Dim value As Texture  
  
value = instance.GetModelTexture(ConfigName)
```

C#

```
Texture GetModelTexture(  
    System.string ConfigName  
)
```

C++/CLI

```
Texture^ GetModelTexture(  
& System.String^ ConfigName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigName
Name of the configuration

Return Value

[ITexture](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetModelTexture](#).

Example

[Get Model Textures \(VBA\)](#)
[Get Model Textures \(VB.NET\)](#)
[Get Model Textures \(C#\)](#)

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetTexture Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

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GetParent Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetParent Method (IComponent2)

Gets the parent component.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetParent() As Component2
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As Component2  
  
value = instance.GetParent()
```

C#

```
Component2 GetParent()
```

C++/CLI

```
Component2^ GetParent();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Parent [component](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetParent](#).

Example

[Get Parent Component \(VBA\)](#)
[Show Selected Components Only \(VBA\)](#)

Remarks

If the component is the root component or a top-level component, then this method returns NULL.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::GetComponentCount Method](#)
[IAssemblyDoc::GetComponents Method](#)
[IComponent2::GetChildren Method](#)
[IComponent2::IGetChildren Method](#)
[IConfiguration::GetRootComponent3 Method](#)
[IConfiguration::IGetRootComponent2 Method](#)
[IConfigurationManager::ActiveConfiguration Property](#)

Availability

SOLIDWORKS 2001Plus SP5, Revision Number 10.5

SOLIDWORKS API Help

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GetPathName Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetPathName Method (IComponent2)

Gets the full path name for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPathName() As System.String
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.String

value = instance.GetPathName()
```

C#

```
System.string GetPathName()
```

C++/CLI

```
System.String^ GetPathName();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Full path name for this component, including the filename

Visual Basic for Applications (VBA) Syntax

See [Component2::GetPathName](#).

Example

[Get Component for Selected Entity \(VBA\)](#)
[Get Components Properties in Drawing View \(VBA\)](#)
[Get Part for Corresponding Component \(VBA\)](#)
[Make Assembly Components Lightweight \(VBA\)](#)
[Set All Assembly Components Lightweight or Resolved \(VBA\)](#)
[Keep SOLIDWORKS Invisible While Activating Documents \(C#\)](#)
[Keep SOLIDWORKS Invisible While Activating Documents \(VB.NET\)](#)
[Keep SOLIDWORKS Invisible While Activating Documents \(VBA\)](#)

Remarks

The underlying document for this component might not have been loaded into memory if the component is lightweight or suppressed. In this situation, [IComponent2::GetModelDoc](#) or [IComponent2::IGetModelDoc](#) may return NULL and the file and path name returned by this method will be the As-Saved file and path name for the IComponent2.

This method does not apply search criteria or look in the current working directory for the component file reference if the component is lightweight or suppressed.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IModelDoc2::GetPathName Method \(\)](#)
[IComponent2::GetImportedPath Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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GetPLMID Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetPLMID Method (IComponent2)

Gets the ID of this [SOLIDWORKS Design connected to the 3DEXPERIENCE platform](#) document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPLMID() As System.String
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.String

value = instance.GetPLMID()
```

C#

```
System.string GetPLMID()
```

C++/CLI

```
System.String^ GetPLMID();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

ID of this document

Visual Basic for Applications (VBA) Syntax

See [Component2::GetPLMID](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2024 SP01, Revision Number 32.1

SOLIDWORKS API Help

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GetPropertyExtension Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetPropertyExtension Method (IComponent2)

Gets the property extension on this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPropertyExtension( _
    ByVal ID As System.Integer _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim ID As System.Integer
Dim value As System.Object

value = instance.GetPropertyExtension(ID)
```

C#

```
System.object GetPropertyExtension(
    System.int ID
)
```

C++/CLI

```
System.Object^ GetPropertyExtension(  
& System.int ID  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID
0

Return Value

Value of the property extension

Visual Basic for Applications (VBA) Syntax

See [Component2::GetPropertyExtension](#).

Remarks

SOLIDWORKS recommends that you use the [IAttribute](#), [IAttributeDef](#), and [IParameter](#) interfaces instead of this method. These interfaces provide more flexibility.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::AddPropertyExtension Method](#)
[IComponent2::ResetPropertyExtension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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GetReferencedDisplayStates Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetReferencedDisplayStates Method (IComponent2)

Gets the display states of this component that are referenced by the specified assembly display state(s).

.NET Syntax

Visual Basic (Declaration)

```
Function GetReferencedDisplayStates( _  
    ByVal AssemblyDisplayStateOption As System.Integer, _  
    ByVal AssemblyDisplayStateNames As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim AssemblyDisplayStateOption As System.Integer  
Dim AssemblyDisplayStateNames As System.Object  
Dim value As System.Object
```

```
value = instance.GetReferencedDisplayStates(AssemblyDisplayStateOption, AssemblyDisplayStateNames)
```

C#

```
System.object GetReferencedDisplayStates(  
    System.int AssemblyDisplayStateOption,  
    System.object AssemblyDisplayStateNames  
)
```

C++/CLI

```
System.Object^ GetReferencedDisplayStates(
& System.int AssemblyDisplayStateOption,
& System.Object^ AssemblyDisplayStateNames
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AssemblyDisplayStateOption

Display state option as defined in [swDisplayStateOpts_e](#)

AssemblyDisplayStateNames

Array of assembly display state names; valid only if AssemblyDisplayStateOption is set to swDisplayStateOpts_e.swSpecifyDisplayState

Return Value

Array of referenced display state names

Visual Basic for Applications (VBA) Syntax

See [Component2::GetReferencedDisplayStates](#).

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

[IComponent2::SetReferencedDisplayStates Method \(\)](#)

[IComponent2::GetVisibilityInAsmDisplayStates Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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GetRenderMaterials Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetRenderMaterials Method (IComponent2)

Obsolete. Superseded by [IComponent2::GetRenderMaterials2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetRenderMaterials() As System.Object
```

Visual Basic (Usage)

```
Dim instance As IComponent2
Dim value As System.Object

value = instance.GetRenderMaterials()
```

C#

```
System.object GetRenderMaterials()
```

C++/CLI

```
System.Object^ GetRenderMaterials();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [appearances](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetRenderMaterials](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IGetRenderMaterials Method](#)
[IComponent2::GetRenderMaterialsCount Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

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GetRenderMaterials2 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetRenderMaterials2 Method (IComponent2)

Gets the appearances applied to this component in the specified display states.

.NET Syntax

Visual Basic (Declaration)

```
Function GetRenderMaterials2( _  
    ByVal DisplayStateOption As System.Integer, _  
    ByVal DisplayStateNames As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim DisplayStateOption As System.Integer  
Dim DisplayStateNames As System.Object  
Dim value As System.Object  
  
value = instance.GetRenderMaterials2(DisplayStateOption, DisplayStateNames)
```

C#

```
System.object GetRenderMaterials2(  
    System.int DisplayStateOption,  
    System.object DisplayStateNames  
)
```

C++/CLI

```
System.Object^ GetRenderMaterials2(  
& System.int DisplayStateOption,  
& System.Object^ DisplayStateNames  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DisplayStateOption
Display states as defined in [swDisplayStateOpts_e](#)
DisplayStateNames
Array of display state names

Return Value

Array of [appearances](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetRenderMaterials2](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

[IComponent2::GetRenderMaterialsCount2 Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

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GetRenderMaterialsCount Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetRenderMaterialsCount Method (IComponent2)

Obsolete. Superseded by [IComponent2::GetRenderMaterialsCount2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetRenderMaterialsCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Integer
```

```
value = instance.GetRenderMaterialsCount()
```

C#

```
System.int GetRenderMaterialsCount()
```

C++/CLI

```
System.int GetRenderMaterialsCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of appearances

Visual Basic for Applications (VBA) Syntax

See [Component2::GetRenderMaterialsCount](#).

Remarks

Call this method before calling [IComponent::IGetRenderMaterials](#) to get the size of the array for that method.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetRenderMaterials Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

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GetRenderMaterialsCount2 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetRenderMaterialsCount2 Method (IComponent2)

Gets the number of appearances applied to this component in the specified display states.

.NET Syntax

Visual Basic (Declaration)

```
Function GetRenderMaterialsCount2( _  
    ByVal DisplayStateOption As System.Integer, _  
    ByVal DisplayStateNames As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim DisplayStateOption As System.Integer  
Dim DisplayStateNames As System.Object  
Dim value As System.Integer  
  
value = instance.GetRenderMaterialsCount2(DisplayStateOption, DisplayStateNames)
```

C#

```
System.int GetRenderMaterialsCount2(  
    System.int DisplayStateOption,  
    System.object DisplayStateNames  
)
```

C++/CLI

```
System.int GetRenderMaterialsCount2(  
& System.int DisplayStateOption,  
& System.Object^ DisplayStateNames  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DisplayStateOption

Display states as defined in [swDisplayStateOpts_e](#)

DisplayStateNames

Array of display state names

Return Value

Number of appearances

Visual Basic for Applications (VBA) Syntax

See [Component2::GetRenderMaterialsCount2](#).

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

[IComponent2::GetRenderMaterials2 Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

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GetRepresentationParent Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetRepresentationParent Method (IComponent2)

Gets the representation parent of this [SOLIDWORKS Design connected to the 3DEXPERIENCE platform](#) document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetRepresentationParent() As System.String
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.String

value = instance.GetRepresentationParent()
```

C#

```
System.string GetRepresentationParent()
```

C++/CLI

```
System.String^ GetRepresentationParent();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Representation parent of this SOLIDWORKS Design connected to the 3DEXPERIENCE platform document

Visual Basic for Applications (VBA) Syntax

See [Component2::GetRepresentationParent](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2024 SP01, Revision Number 32.1

SOLIDWORKS API Help

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GetSectionedBodies Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetSectionedBodies Method (IComponent2)

Gets the sectioned bodies in the specified section view.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSectionedBodies( _
    ByVal ViewIn As System.Object _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim ViewIn As System.Object
Dim value As System.Object

value = instance.GetSectionedBodies(ViewIn)
```

C#

```
System.object GetSectionedBodies(
    System.object ViewIn
)
```

C++/CLI

```
System.Object^ GetSectionedBodies(
    & System.Object^ ViewIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewIn

Model view displaying the sectioned view

Return Value

Array of the sectioned bodies in the specified view

Visual Basic for Applications (VBA) Syntax

See [Component2::GetSectionedBodies](#).

Example

[Get Sectioned Bodies \(VBA\)](#)

Remarks

To determine if the model view you want is currently displaying a sectioned view, use [IModelView::GetDisplayState](#).

If you need the full body representation, use [IComponent2::GetBody](#), [IPartDoc::GetBodies2](#), or [IPartDoc::EnumBodies3](#), which ignore the sectioning operation.

For COM implementations, see [IComponent2::EnumSectionedBodies](#).

If a component is suppressed or lightweight, this method returns NULL because the component is not loaded into memory by SOLIDWORKS. For more information on lightweight components, see [Work With Lightweight Components](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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GetSelectByIdString Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetSelectByIdString Method (IComponent2)

Gets the name of the component for possible use with [IModelDocExtension::SelectById2](#), for selectively opening a document using [ISldWorks::OpenDoc7](#) and [IDocumentSpecification](#), etc.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSelectByIdString() As System.String
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.String

value = instance.GetSelectByIdString()
```

C#

```
System.string GetSelectByIdString()
```

C++/CLI

```
System.String^ GetSelectByIdString();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Name of component

Visual Basic for Applications (VBA) Syntax

See [Component2::GetSelectByIDString](#).

Example

[Get Component Name from Selected Entity \(VB.NET\)](#)
[Get Component Name From Selected Entity \(VBA\)](#)
[Get Component Name from Selected Entity \(C#\)](#)

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::GetComponentByName Method](#)
[IComponent2::Name2 Property](#)
[IAssemblyDoc::GetUnloadedComponentNames Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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GetSmartComponentData Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetSmartComponentData Method (IComponent2)

Gets the features, components, and feature references of a Smart Component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSmartComponentData( _  
    ByRef Features As System.Object, _  
    ByRef FeaturesSelected As System.Object, _  
    ByRef Components As System.Object, _  
    ByRef ComponentsSelected As System.Object, _  
    ByRef References As System.Object _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Features As System.Object  
Dim FeaturesSelected As System.Object  
Dim Components As System.Object  
Dim ComponentsSelected As System.Object  
Dim References As System.Object  
Dim value As System.Boolean
```

```
value = instance.GetSmartComponentData(Features, FeaturesSelected, Components, ComponentsSelected,  
References)
```

C#

```
System.bool GetSmartComponentData(  
    out System.object Features,  
    out System.object FeaturesSelected,  
    out System.object Components,  
    out System.object ComponentsSelected,  
    out System.object References  
)
```

C++/CLI

```

System.bool GetSmartComponentData(
& [Out] System.Object^ Features,
& [Out] System.Object^ FeaturesSelected,
& [Out] System.Object^ Components,
& [Out] System.Object^ ComponentsSelected,
& [Out] System.Object^ References
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Features

Array of [IFeatures](#) that can be selected for this Smart Component (see **Remarks**)

FeaturesSelected

Array of boolean values; true if the corresponding item in the Features array is selected, false if it is not selected

Components

Array of [IComponent2](#)s that can be selected for this Smart Component (see **Remarks**)

ComponentsSelected

Array of boolean values; true if the corresponding item in the Components array is selected, false if it is not selected

References

Array of feature reference entities that are used to activate the Smart Features specified by Features and FeaturesSelected; empty if the Smart Component features have not been activated (see **Remarks**)

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::GetSmartComponentData](#).

Example

[Activate Smart Features in an Assembly \(VBA\)](#)
[Activate Smart Features in an Assembly \(VB.NET\)](#)
[Activate Smart Features in an Assembly \(C#\)](#)

Remarks

A Smart Component is defined by:

- Components
- Features
- Feature references

See the SOLIDWORKS Help for more information about Smart Components.

Use this method to get the current features, components, and feature references of a Smart Component.

Before calling this method:

1. Open an assembly that contains a Smart Component whose Smart Features are not activated.
2. Find the Smart Component in the assembly using [IComponent2::IsSmartComponent](#).

After calling this method, use [IComponent2::SetSmartComponentData](#) to change which features and components of a Smart Component to enable and which Smart Features to activate.

Use [ISmartComponentFeatureData](#) to:

- Insert new features and components into a Smart Component.
- Delete features and components from a Smart Component.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::AddSmartComponent Method](#)
[IAssemblyDoc::CreateSmartComponent Method](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

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GetSpecificTransform Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetSpecificTransform Method (IComponent2)

Get the collapsed or exploded transform of a component when the assembly is exploded.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSpecificTransform( _  
    ByVal IgnoreExplode As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim IgnoreExplode As System.Boolean  
Dim value As System.Object  
  
value = instance.GetSpecificTransform(IgnoreExplode)
```

C#

```
System.object GetSpecificTransform(  
    System.bool IgnoreExplode  
)
```

C++/CLI

```
System.Object^ GetSpecificTransform(  
&    System.bool IgnoreExplode  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IgnoreExplode

True to get the component's collapsed transform when the assembly is exploded, false to get the component's exploded transform when the assembly is exploded

Return Value

[Transform](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetSpecificTransform](#).

Example

[Get Collapsed Transform of Component in Exploded View \(C#\)](#)
[Get Collapsed Transform of Component in Exploded View \(VB.NET\)](#)
[Get Collapsed Transform of Component in Exploded View \(VBA\)](#)

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetTotalTransform Method \(\)](#)
[IComponent2::SetTransformAndSolve2 Method \(\)](#)
[IComponent2::PresentationTransform Property \(\)](#)
[IComponent2::Transform2 Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

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GetSuppression Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetSuppression Method (IComponent2)

Obsolete. Superseded by [IComponent2::GetSuppression2](#).

.NET Syntax

```
Function GetSuppression() As System.Integer
```



```
Dim instance As IComponent2
Dim value As System.Integer

value = instance.GetSuppression()
```

```
System.int GetSuppression()
```

```
System.int GetSuppression();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Suppression state of this component instance as defined in [swComponentSuppressionState](#) e

Visual Basic for Applications (VBA) Syntax

See [Component2::GetSuppression](#).

Example

[Access Assembly \(C++\)](#)
[Change Material Properties \(VBA\)](#)
[Make Assembly Components Lightweight \(VBA\)](#)
[Get Component State \(C#\)](#)
[Get Component State \(VB.NET\)](#)
[Get Component State \(VBA\)](#)

Remarks

Use this method to determine if the component is suppressed, lightweight, or fully resolved. It is critical to know the component's suppression state because lightweight and suppressed components contain only a small subset of data compared to a fully resolved component. For more information, see [Work With Lightweight Components](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::LightweightAllResolved Method](#)
[IAssemblyDoc::SetComponentState Method](#)
[IAssemblyDoc::SetComponentSuppression Method](#)
[IComponent2::IsSuppressed Method](#)
[IComponent2::SetSuppression2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

GetSuppression2 Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetSuppression2 Method (IComponent2)

Gets the suppression state of this component.

.NET Syntax

```
Function GetSuppression2() As System.Integer
```



```
Dim instance As IComponent2
Dim value As System.Integer

value = instance.GetSuppression2()
```

C#

```
System.int GetSuppression2()
```

C++/CLI

```
System.int GetSuppression2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Suppression state of this component or internal ID mismatch error code as defined in [swComponentSuppressionState_e](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetSuppression2](#).

Example

[Get and Set Component's Suppression State \(VBA\)](#)

Remarks

The difference between this method and the now obsolete `IComponent2::GetSuppression` is that this method returns an error code (5 = `swComponentSuppressionState_e.swComponentInternalIdMismatch`) if there is an internal ID mismatch.

Use this method to determine if the component is suppressed, lightweight, or fully resolved. It is critical to know the component's suppression state because lightweight and suppressed components contain only a small subset of data compared to a fully resolved component. For more information, see [Work With Lightweight Components](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::LightweightAllResolved Method \(\)](#)
[IAssemblyDoc::SetComponentState Method \(\)](#)
[IAssemblyDoc::SetComponentSuppression Method \(\)](#)
[IComponent2::IsSuppressed Method \(\)](#)
[IComponent2::SetSuppression2 Method \(\)](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessNorms Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetTessNorms Method (IComponent2)

Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for the component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessNorms() As System.Object
```

Visual Basic (Usage)

```
Dim instance As IComponent2
Dim value As System.Object
```



```
value = instance.GetTessNorms()
```

C#

```
System.object GetTessNorms()
```

C++/CLI

```
System.Object^ GetTessNorms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Object of type SafeArray of floats (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetTessNorms](#).

Remarks

Tessellation information is available only when the component is loaded as lightweight.

The format of retval is:

- float x, y, z - first point's unit normal
- float x, y, z - second point's unit normal
- float x, y, z - third point's unit normal

for the set of triangles for the component.

The total size of the data is [9 x sizeof(float) x (number of triangles)].

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetTessNorms Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessTriangles Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetTessTriangles Method (IComponent2)

Gets the triangles that make up the shaded picture tessellation for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTriangles( _  
    ByVal NoConversion As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IComponent2  
Dim NoConversion As System.Boolean  
Dim value As System.Object
```

```
value = instance.GetTessTriangles(NoConversion)
```

C#



```
System.Object GetTessTriangles(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.Object^ GetTessTriangles(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

True prohibits conversion to user units from system units, false does not

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetTessTriangles](#).

Remarks

Tessellation information is available only when the component is loaded as lightweight.

These triangles are intended for graphics display purposes and do not represent a tessellation that could be used, for example, by a machining application. If you need the type of accuracy associated with a machining product, we recommend that you traverse the body faces and extract the topology and geometry data to create your own faceting.

The format of the returned data is:

- float x, y, z - first point in meters
- float x, y, z - second point in meters
- float x, y, z - third point in meters

for the set of triangles for the component.

The total size of the data is $\lceil 9 \times \text{sizeof}(\text{float}) \times (\text{number of triangles}) \rceil$.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetTessTriangles Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessTriStripEdges Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetTessTriStripEdges Method (IComponent2)

Gets the edge IDs for the triangle strips.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetTessTriStripEdges() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Object

value = instance.GetTessTriStripEdges()
```

C#

```
System.Object GetTessTriStripEdges()
```

C++/CLI

```
System.Object^ GetTessTriStripEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array containing the list of edge IDs

Visual Basic for Applications (VBA) Syntax

See [Component2::GetTessTriStripEdges](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IGetTessTriStripEdges Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessTriStripNorms Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetTessTriStripNorms Method (IComponent2)

Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessTriStripNorms() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Object

value = instance.GetTessTriStripNorms()
```

C#

```
System.Object GetTessTriStripNorms()
```

C++/CLI

```
System.Object^ GetTessTriStripNorms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array containing the tri-strip normals

Visual Basic for Applications (VBA) Syntax

See [Component2::GetTessTriStripNorms](#).

Remarks

Tessellation information is available only when the component is loaded as lightweight.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IGetTessTriStripNorms Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTessTriStrips Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetTessTriStrips Method (IComponent2)

Gets the vertices that make up the shaded picture tessellation for this component.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetTessTriStrips( _  
    ByVal NoConversion As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim NoConversion As System.Boolean  
Dim value As System.Object
```

```
value = instance.GetTessTriStrips(NoConversion)
```

C#

```
System.object GetTessTriStrips(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.Object^ GetTessTriStrips(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion
True prohibits conversion to user units from system units, false does not

Return Value

Array (see [Remarks](#))

Visual Basic for Applications (VBA) Syntax

See [Component2::GetTessTriStrips](#).

Remarks

Tessellation information is available only when the component is loaded as lightweight.

These triangles are intended for graphical display purposes and do not represent a tessellation that could be used, for example, by a machining application. If you need the type of accuracy associated with a machining product, you should traverse the body faces and extract the topology and geometry data to create your own faceting.

where:

- *FaceCount* = number of faces on the body
- *StripCount* = total number of strips on the body
- *VertexCount* x 3 = total number of vertices. Multiplied by three to cover X, Y, and Z
- *NumStrips* = number of strips on a particular face
- *VertexPerStrip* = an array containing the number of vertex points on particular face strip
- *VertexPoints* = an array of X,Y,Z points for each vertex on the particular face strip

Because the returned array elements are of type float, and *FaceCount*, *StripCount*, *VertexCount*, *NumStrips*, and the elements of *VertexPerStrip* are integers that have been packed into the return array elements, and you must [unpack](#) them before you can use them.

For example:

- *sa*[0] = number of faces
- *sa*[1] = total number of strips
- *sa*[2] = total number of vertices x 3 (i.e., - data values)

For each face:

- *sa*[3] = number of strips in this face
- *sa*[4] = number of vertices in first strip of this face
- *sa*[5] = number of vertices in second strip of this face
- ...
- *sa*[3 + *sa*[3]&cd;] = number of vertices in last strip of this face

For each strip, i:

- *sa*[3 + *sa*[3] + 1] = X value of first vertex in this strip of the face
- *sa*[3 + *sa*[3] + 2] = Y value of first vertex in this strip of the face
- *sa*[3 + *sa*[3] + 3] = Z value of first vertex in this strip of the face
- *sa*[3 + *sa*[3] + 4] = X value of second vertex in this strip of the face
- ...
- *sa*[3 + *sa*[3] + (*sa*[3+i] x 3)] = Z value of last vertex in this strip of the face

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IGetTessTriStrips Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTexture Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetTexture Method (IComponent2)

Gets the texture applied to this component in the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTexture( _  
    ByVal Config\_name As System.String _  
) As Texture
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Config_name As System.String  
Dim value As Texture  
  
value = instance.GetTexture(Config_name)
```

C#

```
Texture GetTexture(  
    System.string Config\_name
```

)

C++/CLI

```
Texture^ GetTexture(  
& System.String^ Config\_name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_name
Name of the configuration

Return Value

If texture is...	Then retval is...
Applied to this component in Config_name	ITexture object
Not present	NULL

Visual Basic for Applications (VBA) Syntax

See [Component2::GetTexture](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::RemoveTexture Method](#)
[IComponent2::SetTexture Method](#)
[IComponent2::SetTextureByDisplayState Method](#)
[IComponent2::RemoveTextureByDisplayState Method](#)
[IModelDocExtension::CreateTexture Method](#)
[IComponent2::GetModelTexture Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetTotalTransform Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetTotalTransform Method (IComponent2)

Combines the original transform of this component with the presentation transform of this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTotalTransform(  
    ByVal IncludePresentationXform As System.Boolean _  
) As MathTransform
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim IncludePresentationXform As System.Boolean  
Dim value As MathTransform  
  
value = instance.GetTotalTransform(IncludePresentationXform)
```

C#

```
MathTransform GetTotalTransform(  
    System.bool IncludePresentationXform  
)
```

```
MathTransform^ GetTotalTransform(
& System.bool IncludePresentationXform
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IncludePresentationXform

True to combine the transforms, false to not

Return Value

Pointer to the [IMathTransform](#) object

Visual Basic for Applications (VBA) Syntax

See [Component2::GetTotalTransform](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[AssemblyDoc::EnablePresentation Property](#)
[IComponent2::PresentationTransform Property](#)
[IComponent2::RemovePresentationTransform Method](#)
[IComponent2::SetTransformAndSolve2 Method](#)
[IComponent2::Transform2 Property](#)
[IComponent2::GetSpecificTransform Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetType Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetType Method (IComponent2)

Gets this lightweight assembly component's document type.

.NET Syntax

Visual Basic (Declaration)

```
Function GetType() As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IComponent2
Dim value As System.Integer

value = instance.GetType()
```

C#

```
System.int GetType()
```

C++/CLI

```
System.int GetType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Type of document as defined by [swDocumentTypes_e](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetType](#).



Copy Code

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2024 SP01, Revision Number 32.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUnloadedComponentNames Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetUnloadedComponentNames Method (IComponent2)

Gets the component's unloaded children components' path names, referenced configuration names, reasons why they are unloaded, document types, and names.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUnloadedComponentNames( _  
    ByRef UnloadedComponentPathNames As System.Object, _  
    ByRef UnloadedComponentReferencedConfigurationNames As System.Object, _  
    ByRef ReasonForUnloadingComponents As System.Object, _  
    ByRef DocTypes As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim UnloadedComponentPathNames As System.Object  
Dim UnloadedComponentReferencedConfigurationNames As System.Object  
Dim ReasonForUnloadingComponents As System.Object  
Dim DocTypes As System.Object  
Dim value As System.Object  
  
value = instance.GetUnloadedComponentNames(UnloadedComponentPathNames,  
UnloadedComponentReferencedConfigurationNames, ReasonForUnloadingComponents, DocTypes)
```

C#

```
System.object GetUnloadedComponentNames(  
    out System.object UnloadedComponentPathNames,  
    out System.object UnloadedComponentReferencedConfigurationNames,  
    out System.object ReasonForUnloadingComponents,  
    out System.object DocTypes  
)
```

C++/CLI

```
System.Object^ GetUnloadedComponentNames(  
& [Out] System.Object^ UnloadedComponentPathNames,  
& [Out] System.Object^ UnloadedComponentReferencedConfigurationNames,  
& [Out] System.Object^ ReasonForUnloadingComponents,  
& [Out] System.Object^ DocTypes  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UnloadedComponentPathNames
Array of children components' path names
UnloadedComponentReferencedConfigurationNames
Array of children components' referenced configuration names
ReasonForUnloadingComponents
Array indicating the reason each child component is unloaded as defined by [swComponentLoadStatus_e](#)
DocTypes
Array of document types as defined in [swDocumentTypes_e](#)

Return Value

Array of children components' names

Visual Basic for Applications (VBA) Syntax

See [Component2::GetUnloadedComponentNames](#).

Remarks

This method is useful when the assembly document was opened with [Quick View/Selective open](#).

To get whether a component has hidden or suppressed children components, call [IComponent2::HasUnloadedComponents](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetHiddenUnloadedChildrenCount Method](#)
[IComponent2::HasUnloadedComponents Method](#)
[IAssemblyDoc::GetUnloadedComponentNames Method](#)
[IAssemblyDoc::HasUnloadedComponents Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetVisibility Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetVisibility Method (IComponent2)

Gets the visibility state for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetVisibility( _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_names As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Config_opt As System.Integer  
Dim Config_names As System.Object  
Dim value As System.Object  
  
value = instance.GetVisibility(Config_opt, Config_names)
```

C#

```
System.object GetVisibility(  
    System.int Config\_opt,  
    System.object Config\_names  
)
```

C++/CLI

```
System.Object^ GetVisibility(  
& System.int Config\_opt,  
& System.Object^ Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt
Configuration option as defined in [swInConfigurationOpts_e](#)

Config_names

Array of the configuration names for this component; valid only if Config_opt is set to swInConfigurationOpts_e.swSpecifyConfiguration

Return Value

Array of the visibility states for this component

Visual Basic for Applications (VBA) Syntax

See [Component2::GetVisibility](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetVisibility Method](#)
[IComponent2::SetVisibility Method](#)
[IComponent2::GetVisibility Method](#)
[IComponent2::Visible Property](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetVisibilityInAsmDisplayStates Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetVisibilityInAsmDisplayStates Method (IComponent2)

Gets the visibilities of this component in the specified assembly display state(s).

.NET Syntax

Visual Basic (Declaration)

```
Function GetVisibilityInAsmDisplayStates( _  
    ByVal AssemblyDisplayStateOption As System.Integer, _  
    ByVal AssemblyDisplayStateNames As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim AssemblyDisplayStateOption As System.Integer  
Dim AssemblyDisplayStateNames As System.Object  
Dim value As System.Object  
  
value = instance.GetVisibilityInAsmDisplayStates(AssemblyDisplayStateOption,  
AssemblyDisplayStateNames)
```

C#

```
System.object GetVisibilityInAsmDisplayStates(  
    System.int AssemblyDisplayStateOption,  
    System.object AssemblyDisplayStateNames  
)
```

C++/CLI

```
System.Object^ GetVisibilityInAsmDisplayStates(  
& System.int AssemblyDisplayStateOption,  
& System.Object^ AssemblyDisplayStateNames  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AssemblyDisplayStateOption

Display state option as defined in [swDisplayStateOpts_e](#)

AssemblyDisplayStateNames

Array of assembly display state names; valid only if AssemblyDisplayStateOption is set to swDisplayStateOpts_e.swSpecifyDisplayState

Return Value

Array of visibilities as defined in [swComponentVisibilityState](#) ^e

 Visual Basic for Applications (VBA) Syntax

See [Component2::GetVisibilityInAsmDisplayStates](#).

 See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetReferencedDisplayStates Method \(\)](#)
[IComponent2::SetVisibilityInAsmDisplayStates Method \(\)](#)

 Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetXform Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetXform Method (IComponent2)

Obsolete. Superseded by [IComponent2::Transform2](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function GetXform() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Object  
  
value = instance.GetXform()
```

C#

```
System.object GetXform()
```

C++/CLI

```
System.Object^ GetXform();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

 Visual Basic for Applications (VBA) Syntax

See [Component2::GetXform](#).

 See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

HasAssemblyFeatures Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : HasAssemblyFeatures Method (IComponent2)

Gets whether this component has assembly features in the specified configuration.

 .NET Syntax

Visual Basic (Declaration)

```
Function HasAssemblyFeatures(  
    ByRef ConfigurationName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim ConfigurationName As System.String  
Dim value As System.Boolean  
  
value = instance.HasAssemblyFeatures(ConfigurationName)
```

C#

```
System.bool HasAssemblyFeatures(  
    out System.string ConfigurationName  
)
```

C++/CLI

```
System.bool HasAssemblyFeatures(  
& [Out] System.String^ ConfigurationName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigurationName
Configuration name

Return Value

True if has assembly features, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::HasAssemblyFeatures](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2026 SP02, Revision Number 34.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

HasMaterialPropertyValues Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : HasMaterialPropertyValues Method (IComponent2)

Gets whether this component has an appearance.

.NET Syntax

Visual Basic (Declaration)

```
Function HasMaterialPropertyValues() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean  
  
value = instance.HasMaterialPropertyValues()
```

C#

```
System.bool HasMaterialPropertyValues()
```

C++/CLI

```
System.bool HasMaterialPropertyValues();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this component has an appearance, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::HasMaterialPropertyValues](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::MaterialPropertyValues Property \(\)](#)
[IComponent2::GetMaterialPropertyValues2 Method \(\)](#)
[IComponent2::RemoveMaterialProperty2 Method \(\)](#)
[IComponent2::GetModelMaterialPropertyValues Method \(\)](#)
[IComponent2::SetMaterialPropertyValues2 Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

HasUnloadedComponents Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : HasUnloadedComponents Method (IComponent2)

Gets whether this component has hidden or suppressed unloaded children components.

.NET Syntax

Visual Basic (Declaration)

```
Function HasUnloadedComponents() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean  
  
value = instance.HasUnloadedComponents()
```

C#

```
System.bool HasUnloadedComponents()
```

C++/CLI

```
System.bool HasUnloadedComponents();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this component has hidden or suppressed unloaded children components, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::HasUnloadedComponents](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetUnloadedComponentNames Method](#)
[IComponent2::GetHiddenUnloadedChildrenCount Method](#)
[IAssemblyDoc::HasUnloadedComponents Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IFindAttribute Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IFindAttribute Method (IComponent2)

Finds an attribute on a component.

.NET Syntax

Visual Basic (Declaration)

```
Function IFindAttribute( _  
    ByVal AttributeDef As AttributeDef, _  
    ByVal WhichOne As System.Integer _  
) As Attribute
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim AttributeDef As AttributeDef  
Dim WhichOne As System.Integer  
Dim value As Attribute  
  
value = instance.IFindAttribute(AttributeDef, WhichOne)
```

C#

```
Attribute IFindAttribute(  
    AttributeDef AttributeDef,  
    System.int WhichOne  
)
```

C++/CLI

```
Attribute^ IFindAttribute(  
& AttributeDef^ AttributeDef,  
& System.int WhichOne  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AttributeDef

Pointer to an [attribute definition](#) that you are looking for on this component

WhichOne

Index number of the attribute that you want to return; there might be several attributes on this component

Return Value

Pointer to the [attribute](#)

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::FindAttribute Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetBody Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetBody Method (IComponent2)

Gets the body that belongs to this instance of this component.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetBody() As Body2
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As Body2

value = instance.IGetBody()
```

C#

```
Body2 IGetBody()
```

C++/CLI

```
Body2^ IGetBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IBody2](#) object

Example

[Enumerate Bodies \(C++\)](#)

Remarks

This method returns a valid IBody2 object only for fully resolved components that reference an [IPartDoc](#). For the root component, lightweight components or components that reference an AssemblyDoc, this method returns a NULL body object. However, if a component is in a multibody part, then this method returns the first body.

For COM applications, this method returns the status code ITF_E_COMPONENTNOTRESOLVED.

This method is different from the IPartDoc::Body method in that it recognizes assembly-level features and returns that information based on the component instance. IPartDoc::Body never recognizes assembly-level features because the feature information is kept with the assembly, not propagated down to the part file.

For example, if PartABC is added to an assembly twice, then changes to that IPartDoc object would affect both instances of the assembly IComponent2. Likewise, querying information from that IPartDoc object will not recognize changes in the assembly that may have altered only one of the components (for example, an assembly-level feature was added to one of the components). However, the component object recognizes the two instances of PartABC as two distinct IComponent2 objects and returns its information from the assembly level.

For example, one assembly configuration might have an assembly-level feature that cuts a hole through each of the components in the assembly. Using this method on each of the assembly components returns the body of each component with the hole feature that was applied in this particular configuration. If you switch to the configuration without the assembly-level hole and re-traverse the IComponent2 objects, then calling this method for each component returns the IBody2 object without the hole feature because it was applied in the other configuration.

For more information on lightweight components, see [IComponent2::GetSuppression](#), [IComponent2::SetSuppression2](#), and [IAssemblyDoc::ResolveAllLightWeightComponents](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

[IComponent2::GetBody Method](#)
[IComponent2::GetBodies3 Method](#)
[IComponent2::EnumBodies2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetBox Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetBox Method (IComponent2)

Gets the bounding box for component.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetBox( _  
    ByVal IncludeRefPlanes As System.Boolean, _  
    ByVal IncludeSketches As System.Boolean _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim IncludeRefPlanes As System.Boolean  
Dim IncludeSketches As System.Boolean  
Dim value As System.Double  
  
value = instance.IGetBox(IncludeRefPlanes, IncludeSketches)
```

C#

```
System.double IGetBox(  
    System.bool IncludeRefPlanes,  
    System.bool IncludeSketches  
)
```

C++/CLI

```
System.double IGetBox(  
& System.bool IncludeRefPlanes,  
& System.bool IncludeSketches  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IncludeRefPlanes

True includes reference planes with the returned bounding box, false does not

IncludeSketches

True includes sketches with the returned bounding box, false does not

Return Value

Two diagonal points that bound the component; the format is a pointer to an array of 6 doubles

Remarks

The resulting box encloses the object, but it might not be the tightest box.

The X, Y, Z points returned by SOLIDWORKS are the lower- and upper-diagonal corners that bound the component with the box sides parallel to the X, Y and Z axes. SOLIDWORKS returns box dimensions that enclose the component and are typically close to the minimum possible size.

The return value is an array of doubles as follows:

[*XCorner1*, *YCorner1*, *ZCorner1*, *XCorner2*, *YCorner2*, *ZCorner2*]

It is possible for this method to return S_false for COM or a NULL VARIANT for Dispatch. This occurs if your application calls

IComponent2::IGetBox with a component that represented a subassembly and that subassembly is not loaded. Once the subassembly is loaded, SOLIDWORKS returns the correct bounds and IComponent2::IGetBox returns S_OK.

The user interface behavior is the same. When the user selects a subassembly that is not loaded, there is no selection box around the subassembly. However, once the subassembly is loaded, there is a selection box.

IMPORTANT: The values returned are approximate and should not be used for comparison or calculation purposes. Furthermore, the bounding box may vary after rebuilding the model.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetBox Method](#)
[IModelDocExtension::SetVisibleBox Method](#)
[IModelDocExtension::RemoveVisibleBox Method](#)
[IModelDocExtension::GetVisibleBox Method](#)
[IFeature::GetBox Method](#)
[IFeature::IGetBox Method](#)
[IFace2::GetBox Method](#)
[IFace2::IGetBox Method](#)
[IBody2::GetBodyBox Method](#)
[IBody2::IGetBodyBox Method](#)
[IAssemblyDoc::GetBox Method](#)
[IAssemblyDoc::IGetBox Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetChildren Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetChildren Method (IComponent2)

Gets all of the children components of this component.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetChildren() As Component2
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As Component2

value = instance.IGetChildren()
```

C#

```
Component2 IGetChildren()
```

C++/CLI

```
Component2^ IGetChildren();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of [IComponent2](#) objects
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

If this assembly component is a part document, SOLIDWORKS returns NULL. If this assembly component is the root component or a subassembly, then this method returns the child components that belong to the assembly document.

The typical order of calls needed in assembly traversal is:

1. [IConfigurationManager::ActiveConfiguration](#) (called only once)
2. [IConfiguration::GetRootComponent](#) (called only once)
3. [IComponent2::GetChildren](#) (called recursively)

COM applications should use to [IComponent2::IGetChildrenCount](#) to determine the number of component children, which is the size of the array required to IComponent2::IGetChildren. Because IComponent2::IGetChildren returns an array, this code must be used in an in-process DLL.

You must call the IComponent2::GetChildren method recursively because it returns only the immediate (one level) children. It does not get child components of the sub-assemblies. For example, if one of the child components of a component is a sub-assembly that has its own children, those children are not returned by this method. You need to call this method again from that sub-assembly component to get its children.

For a given component, this method returns all of the immediate child components. This includes suppressed, hidden, and lightweight components. Use [IComponent2::Visible](#) and [IComponent2::GetSuppression](#) to detect the component states.

Results of an assembly traversal vary based on the configuration currently displayed in your main assembly and based on the configuration referenced by the subassembly component. The list of child components that this method returns can be different depending on which configuration is referenced by the component (see IConfigurationManager::ActiveConfiguration and [IComponent2::ReferencedConfiguration](#)).

For example, if one configuration of your main assembly contains a suppressed subassembly, IComponent2::GetChildren returns an empty array when you call it from that suppressed subassembly component. As another example, a subassembly document (.sldasm file) can contain several configurations, each of which has varying states of suppression for its child components. When inserted into your main assembly, this subassembly document can reference any of these configurations. As a result, you might find that the child component suppression states vary based on which configuration is referenced by the subassembly component.

NOTE: Components might not be returned in the same order from call to call.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::GetComponentCount Method](#)
[IAssemblyDoc::IGetComponents Method](#)
[IComponent2::GetParent Method](#)
[IConfigurationManager::ActiveConfiguration Property](#)
[IConfiguration::IGetRootComponent2 Method](#)
[IConfiguration::IGetRootComponent3 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetChildrenCount Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetChildrenCount Method (IComponent2)

Gets the number of children components for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetChildrenCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Integer  
  
value = instance.IGetChildrenCount()
```

C#

```
System.int IGetChildrenCount()
```

C++/CLI

```
System.int IGetChildrenCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of children

Example

[Show All Components \(VBA\)](#)

Remarks

This method returns a single-level count. The [IComponent2::GetChildren](#) and [IComponent2::IGetChildren](#) do not recurse sub-assemblies.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetCorrespondingEntity Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetCorrespondingEntity Method (IComponent2)

Gets the entity that corresponds with the Dispatch pointer in the context of the component.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetCorrespondingEntity( _  
    ByVal PEntity As Entity _  
) As Entity
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim PEntity As Entity  
Dim value As Entity  
  
value = instance.IGetCorrespondingEntity(PEntity)
```

C#

```
Entity IGetCorrespondingEntity(  
    Entity PEntity  
)
```

C++/CLI

```
Entity^ IGetCorrespondingEntity(  
& Entity^ PEntity  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PEntity

Pointer to the [IEntity](#) object (vertex, face, or edge)

Return Value

Pointer to the corresponding object in the context of the component

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetCorresponding Method](#)
[IComponent2::GetCorrespondingEntity Method](#)
[IModelDocExtension::GetCorresponding Method](#)
[IModelDocExtension::GetCorrespondingEntity Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDecals Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetDecals Method (IComponent2)

Gets the decals applied to this component.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDecals( _  
    ByVal Count As System.Integer _  
) As Decal
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Count As System.Integer  
Dim value As Decal  
  
value = instance.IGetDecals(Count)
```

C#

```
Decal IGetDecals(  
    System.int Count  
)
```

C++/CLI

```
Decal^ IGetDecals(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of decals

Return Value

- in-process, unmanaged C++: Pointer to an array of [decals](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IComponent2::GetDecalsCount](#) to get the value of Count.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetDecals Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMaterialPropertyValues2 Method (IComponent2)

[See Also](#)



Gets the material properties for this component.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetMaterialPropertyValues2( _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_count As System.Integer, _  
    ByRef Config\_names As System.String _  
    ) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Config_opt As System.Integer  
Dim Config_count As System.Integer  
Dim Config_names As System.String  
Dim value As System.Double  
  
value = instance.IGetMaterialPropertyValues2(Config_opt, Config_count, Config_names)
```

C#

```
System.double IGetMaterialPropertyValues2(  
    System.int Config\_opt,  
    System.int Config\_count,  
    ref System.string Config\_names  
)
```

C++/CLI

```
System.double IGetMaterialPropertyValues2(  
& System.int Config\_opt,  
& System.int Config\_count,  
& System.String^% Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_count

Number of configurations for this component

Config_names

Array of configuration names for this component (see **Remarks**)

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles of size 9 doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

If you set Config_opt to swThisConfiguration or swAllConfiguration, then pass an empty array because Config_names is ignored.

The format of material_values is an array of doubles:

[*R*, *G*, *B*, *Ambient*, *Diffuse*, *Specular*, *Shininess*, *Transparency*, *Emission*]

Valid values are from 0 to 1 for all variables. If material_values is all -1 values, then material property values were not assigned to the component. Therefore, the component will have the default properties in the user interface.

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

[IComponent2::IRemoveMaterialProperty2 Method](#)

[IComponent2::ISetMaterialPropertyValues2 Method](#)

[IComponent2::IMaterialPropertyValues Property](#)
[IComponent2::GetMaterialPropertyValues2 Method](#)
[IComponent2::HasMaterialPropertyValues Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMaterialPropertyValuesForFace Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetMaterialPropertyValuesForFace Method (IComponent2)

Gets the color of the specified face.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMaterialPropertyValuesForFace( _  
    ByVal FaceIn As System.Object _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim FaceIn As System.Object  
Dim value As System.Double
```

```
value = instance.IGetMaterialPropertyValuesForFace(FaceIn)
```

C#

```
System.double IGetMaterialPropertyValuesForFace(  
    System.object FaceIn  
)
```

C++/CLI

```
System.double IGetMaterialPropertyValuesForFace(  
    & System.Object^ FaceIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceIn

Pointer to the face from which to get its color

Return Value

Color of the face

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2003 SP1, Revision Number 11.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetModelDoc Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetModelDoc Method (IComponent2)

Obsolete. Superseded by [IComponent2::GetModelDoc2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetModelDoc() As ModelDoc2
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As ModelDoc2  
  
value = instance.IGetModelDoc()
```

C#

```
ModelDoc2 IGetModelDoc()
```

C++/CLI

```
ModelDoc2^ IGetModelDoc();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [model document](#)

Remarks

This method might return NULL if:

- a component is suppressed or lightweight.
- the component ID is not loaded into memory by SOLIDWORKS.

For more information on lightweight components, see [Work With Lightweight Components](#).

When you use the IModelDoc2 object of a component, you do not have access to whatever uniqueness might exist for this instance of the assembly IComponent2. This occurs because the IModelDoc2 object goes back to the saved part file. By comparison, the IComponent2 object gathers information at the assembly level. This allows IComponent2 objects to recognize assembly-level changes made to a component instance (for example, assembly-level features and material changes).

In addition, the IModelDoc2 object returned from this method represents the last-saved state. If the component part is currently open, then the IModelDoc2 object represents the state of the opened document. For example, if the component part is not open and it was last saved in the default configuration, then IComponent2::GetModelDoc returns a IModelDoc2 pointer representing that state. To get access to other configuration information (such as features and configuration properties), you must activate the part and show the desired configuration using [IModelDoc2::ShowConfiguration2](#).

If this component is the root component, then this method returns a NULL pointer. For more information, see [IConfiguration::IGetRootComponent2](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetModelDoc Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetModelMaterialPropertyValues Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetModelMaterialPropertyValues Method (IComponent2)

Gets the material properties of this lightweight component in the specified configuration.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetModelMaterialPropertyValues( _  
    ByVal ConfigName As System.String _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim ConfigName As System.String  
Dim value As System.Double  
  
value = instance.IGetModelMaterialPropertyValues(ConfigName)
```

C#

```
System.double IGetModelMaterialPropertyValues(  
    System.string ConfigName  
)
```

C++/CLI

```
System.double IGetModelMaterialPropertyValues(  
& System.String^ ConfigName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigName
Name of the configuration

Return Value

- in-process, unmanaged C++: Pointer to an array of 9 doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

The contents of the returned array:

[*R*, *G*, *B*, *Ambient*, *Diffuse*, *Specularity*, *Shininess*, *Transparency*, *Emission*]

Valid values are between 0.0 and 1.0, inclusive. Multiply the R, G, and B values by 255 to obtain the red, green, and blue component values. Multiply the other values in the array by 100 to obtain percentages. If all values in the array are -1, then material property values were not assigned to this component, and the component has the same default properties as the user interface.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetModelMaterialPropertyValues Method](#)
[IComponent2::GetMaterialPropertyValues2 Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetRenderMaterials Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetRenderMaterials Method (IComponent2)

Obsolete. Superseded by [IComponent2::GetRenderMaterials2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetRenderMaterials( _  
    ByVal Count As System.Integer _  
) As RenderMaterial
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Count As System.Integer  
Dim value As RenderMaterial  
  
value = instance.IGetRenderMaterials(Count)
```

C#

```
RenderMaterial IGetRenderMaterials(  
    System.int Count  
)
```

C++/CLI

```
RenderMaterial^ IGetRenderMaterials(  
&    System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of appearances

Return Value

- in-process, unmanaged C++: Pointer to an array of [appearances](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IComponent2::GetRenderMaterialsCount](#) to get the value of Count.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetRenderMaterials Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

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IGetTemporaryBodyID Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetTemporaryBodyID Method (IComponent2)

Gets the current body tag ID, which is not a permanent ID.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTemporaryBodyID() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Integer
```

```
value = instance.IGetTemporaryBodyID()
```

C#

```
System.int IGetTemporaryBodyID()
```

C++/CLI

```
System.int IGetTemporaryBodyID();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Body ID

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetTessNorms Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetTessNorms Method (IComponent2)

Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for the component.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessNorms() As System.Single
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Single  
  
value = instance.IGetTessNorms()
```

C#

```
System.float IGetTessNorms()
```

C++/CLI

```
System.float IGetTessNorms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of floats (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Tessellation information is available only when the component is loaded as lightweight.

The format of retval is:

- float x, y, z - first point's unit normal
- float x, y, z - second point's unit normal
- float x, y, z - third point's unit normal

for the set of triangles for the component.

The total size of the data is [9 x sizeof(float) x (*number of triangles*)].

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetTessNorms Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTessTriangleCount Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetTessTriangleCount Method (IComponent2)

Gets the number of triangles that make up the shaded picture tessellation for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriangleCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Integer  
  
value = instance.IGetTessTriangleCount()
```

C#

```
System.int IGetTessTriangleCount()
```

C++/CLI

```
System.int IGetTessTriangleCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of triangles that make up the shaded picture tessellation for this component

Remarks

Tessellation information is available only when the component is loaded as lightweight.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTessTriangles Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetTessTriangles Method (IComponent2)

Gets the triangles that make up the shaded picture tessellation for this component.

■ .NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriangles( _  
    ByVal NoConversion As System.Boolean _  
) As System.Single
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim NoConversion As System.Boolean  
Dim value As System.Single  
  
value = instance.IGetTessTriangles(NoConversion)
```

C#

```
System.float IGetTessTriangles(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.float IGetTessTriangles(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

True prohibits conversion to user units from system units, false does not

Return Value

- in-process, unmanaged C++: Pointer to array of floats (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

■ Remarks

Tessellation information is available only when the component is loaded as lightweight.

These triangles are intended for graphics display purposes and do not represent a tessellation that could be used, for example, by a machining application. If you need the type of accuracy associated with a machining product, we recommend that you traverse the body faces and extract the topology and geometry data to create your own faceting.

The format of the returned data is:

- float x, y, z - first point in meters
- float x, y, z - second point in meters
- float x, y, z - third point in meters

for the set of triangles for the component.

The total size of the data is [9 x sizeof(float) x (number of triangles)].

■ See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetTessTriangles Method](#)

■ Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetTessTriStripEdges Method (IComponent2)

Gets the edge IDs for the triangle strips.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriStripEdges() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Integer
```

```
value = instance.IGetTessTriStripEdges()
```

C#

```
System.int IGetTessTriStripEdges()
```

C++/CLI

```
System.int IGetTessTriStripEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of type long containing the list of edge ID
- VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetTessTriStripEdges Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTessTriStripEdgeSize Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetTessTriStripEdgeSize Method (IComponent2)

Gets the number of tessellation triangle edges.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriStripEdgeSize() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Integer
```

```
value = instance.IGetTessTriStripEdgeSize()
```

C#

```
System.int IGetTessTriStripEdgeSize()
```

C++/CLI

```
System.int IGetTessTriStripEdgeSize();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of tessellation triangle edges

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTessTriStripNormals Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetTessTriStripNormals Method (IComponent2)

Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriStripNormals() As System.Single
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Single
```

```
value = instance.IGetTessTriStripNormals()
```

C#

```
System.float IGetTessTriStripNormals()
```

C++/CLI

```
System.float IGetTessTriStripNormals();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of floats containing the tri-strip normals
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

Tessellation information is available only when the component is loaded as lightweight.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IGetTessTriStripNormals Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IGetTessTriStrips Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetTessTriStrips Method (IComponent2)

Gets the vertices that make up the shaded picture tessellation for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriStrips( _  
    ByVal NoConversion As System.Boolean _  
) As System.Single
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim NoConversion As System.Boolean  
Dim value As System.Single  
  
value = instance.IGetTessTriStrips(NoConversion)
```

C#

```
System.float IGetTessTriStrips(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.float IGetTessTriStrips(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

True prohibits conversion to user units from system units, false does not

Return Value

- in-process, unmanaged C++: Pointer to an array of floats (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Tessellation information is available only when the component is loaded as lightweight.

These triangles are intended for graphical display purposes and do not represent a tessellation that could be used, for example, by a machining application. If you need the type of accuracy associated with a machining product, you should traverse the body faces and extract the topology and geometry data to create your own faceting.

where:

- *FaceCount* = number of faces on the body
- *StripCount* = total number of strips on the body
- *VertexCount* x 3 = total number of vertices. Multiplied by three to cover X, Y, and Z
- *NumStrips* = number of strips on a particular face
- *VertexPerStrip* = an array containing the number of vertex points on particular face strip
- *VertexPoints* = an array of X, Y, Z points for each vertex on the particular face strip

Because the returned array elements are of type float, and *FaceCount*, *StripCount*, *VertexCount*, *NumStrips*, and the elements of *VertexPerStrip* are integers that have been packed into the return array elements, and you must [unpack](#) them before you can use them.

For example:

- sa[0] = number of faces
- sa[1] = total number of strips
- sa[2] = total number of vertices x 3 (i.e., - data values)

For each face:

- sa[3] = number of strips in this face
- sa[4] = number of vertices in first strip of this face
- sa[5] = number of vertices in second strip of this face
- ...
- sa[3 + sa[3]&cd;] = number of vertices in last strip of this face

For each strip, i:

- $sa[3 + sa[3] + 1] = X$ value of first vertex in this strip of the face
- $sa[3 + sa[3] + 2] = Y$ value of first vertex in this strip of the face
- $sa[3 + sa[3] + 3] = Z$ value of first vertex in this strip of the face
- $sa[3 + sa[3] + 4] = X$ value of second vertex in this strip of the face
- ...
- $sa[3 + sa[3] + (sa[3+i] \times 3)] = Z$ value of last vertex in this strip of the face

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetTessTriStrips Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTessTriStripSize Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetTessTriStripSize Method (IComponent2)

Gets the array size of floats required to contain the data returned when calling [IComponent2::IGetTessTriStrips](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriStripSize() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Integer  
  
value = instance.IGetTessTriStripSize()
```

C#

```
System.int IGetTessTriStripSize()
```

C++/CLI

```
System.int IGetTessTriStripSize();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Size of the array

Remarks

Tessellation information is available only when the component is loaded as lightweight.

SOLIDWORKS calculates this number as (3 + FaceCount + StripCount + 3 * VertexCount).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

IGetVisibility Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetVisibility Method (IComponent2)

Gets the visibility state for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetVisibility( _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_count As System.Integer, _  
    ByRef Config\_names As System.String _  
    ) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Config_opt As System.Integer  
Dim Config_count As System.Integer  
Dim Config_names As System.String  
Dim value As System.Integer  
  
value = instance.IGetVisibility(Config_opt, Config_count, Config_names)
```

C#

```
System.int IGetVisibility(  
    System.int Config\_opt,  
    System.int Config\_count,  
    ref System.string Config\_names  
)
```

C++/CLI

```
System.int IGetVisibility(  
& System.int Config\_opt,  
& System.int Config\_count,  
& System.String^% Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_count

Number of configurations for this component

Config_names

Array of configuration names of size Config_count

Return Value

Array of visibility states of size Config_count

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

[IComponent2::GetVisibility Method](#)

[IComponent2::SetVisibility Method](#)

[IComponent2::Visible Property](#)

[IAssemblyDoc::SetComponentVisibility Method](#)

Availability

IGetXform Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : IGetXform Method (IComponent2)

Obsolete. Superseded by [IComponent2::Transform2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetXform() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Double  
  
value = instance.IGetXform()
```

C#

```
System.double IGetXform()
```

C++/CLI

```
System.double IGetXform();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)